

Lab 5: Latches and Flip - Flops

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Summary

This report explores the design, simulation and hardware implementation of fundamental sequential logic elements, including SR latches, D latches, D flip-flops, and T flip-flops. By constructing these devices from basic NAND, NOR, and NOT gates, the experiment successfully demonstrates the transition from level sensitive storage to edge triggered synchronization. Through the use of Intel Quartus II software and designed circuits and experimental waveforms used in comparison to theoretical truth tables, the behavior of each part of the lab is able to be understood and learned.

Introduction

Background

Digital memory is built on sequential logic, where feedback loops allow circuits to store a bit of data based on both current and past inputs. This lab focuses on the evolution of these memory elements, starting with the SR latch and D latch, which are level sensitive devices that maintain a state as long as an enable signal is active. To achieve the precise synchronization required for modern computing, these designs are refined into D and T flip-flops which utilize edge triggering to update data only at the specific transition of a clock pulse. Understanding these components makes it essential for designing complex systems like registers and counters, where stable and timed data storage is critical.

Objective of the Report

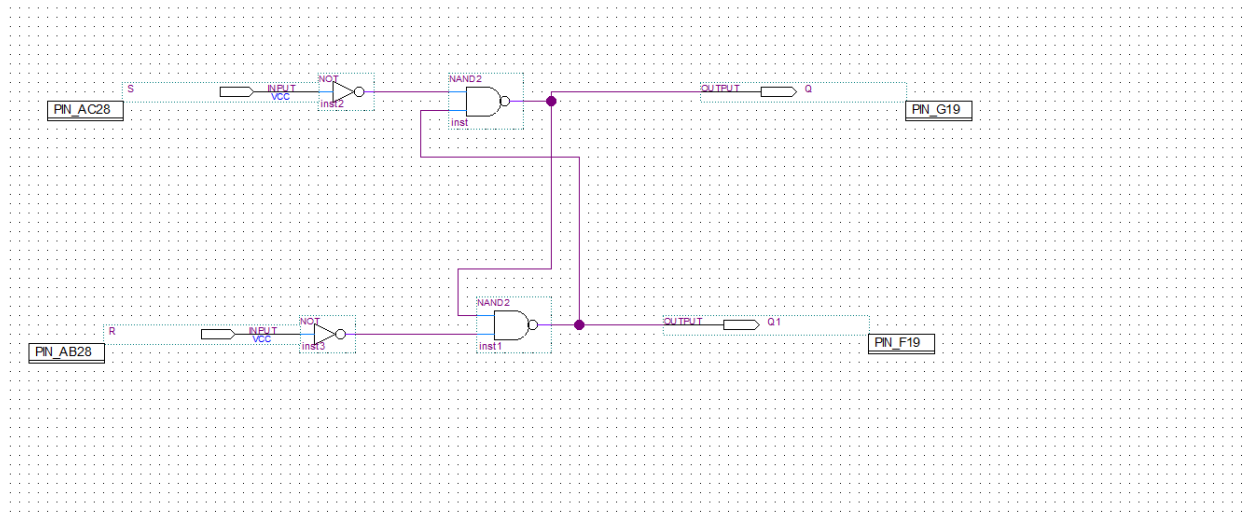
The objective is to compare theoretical and experimental results. Things such as converting boolean expressions into logic circuits for each part of the lab and to find out key characteristics about each device in the lab such as the SR and D latches for example, and how they function.

Pre Lab

The objective of the pre-lab was to study the behaviour of several sequential logic elements and prepare their circuit implementations before beginning the in-lab testing phase. This experiment focused on circuits that are able to store information and whose outputs depend not only on the present inputs but also on the previous state of the system. The pre-lab therefore involved examining the provided function tables, drawing the required block diagrams, and implementing each device using the specified basic gates so that the designs could later be tested in Quartus.

Part 1 - SR Latch

The first part of the pre-lab focused on the SR latch. By observing the function table provided in the lab instructions, the table for the SR latch was prepared and the circuit was implemented using NAND gates. The latch was drawn as a bistable storage element with complementary outputs Q and Q', and its behaviour was analyzed under the set, reset, hold, and forbidden conditions. This helped establish the relationship between the two control inputs and the stored output state before the circuit was recreated in Quartus.

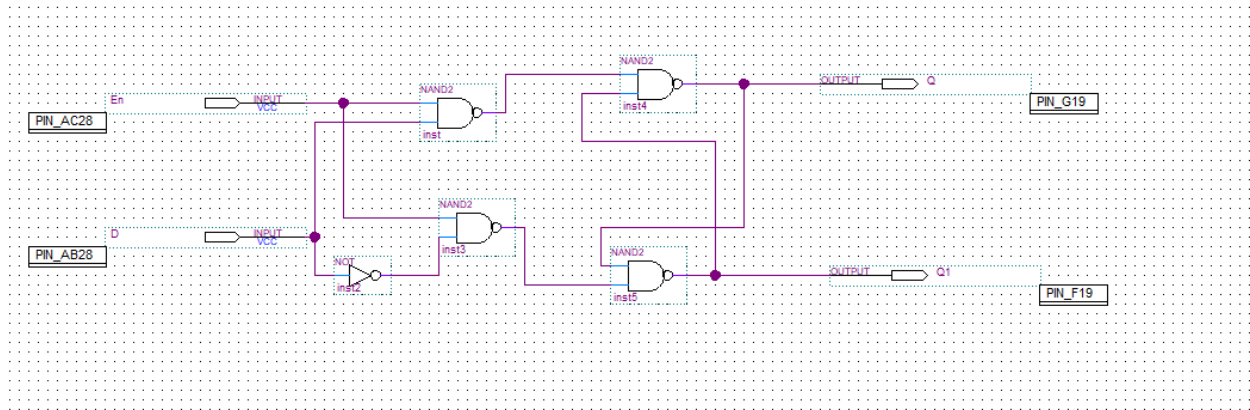


1.

S	R	Q(t)	Q(t+1)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

Part 2- D Latch

The second part of the pre-lab focused on the D latch. This circuit was derived from the SR latch structure in order to eliminate the forbidden state and simplify control of the output through a data input D and an enable input EN. A block diagram of the SR latch was first reviewed and then used as the basis for the D latch implementation using NAND gates. The table for the D latch was also written by analyzing how the output behaves when the enable signal is active and when it is inactive. In this design, when the enable signal is active, the latch output follows the data input, while when the enable signal is inactive, the previously stored value is preserved.

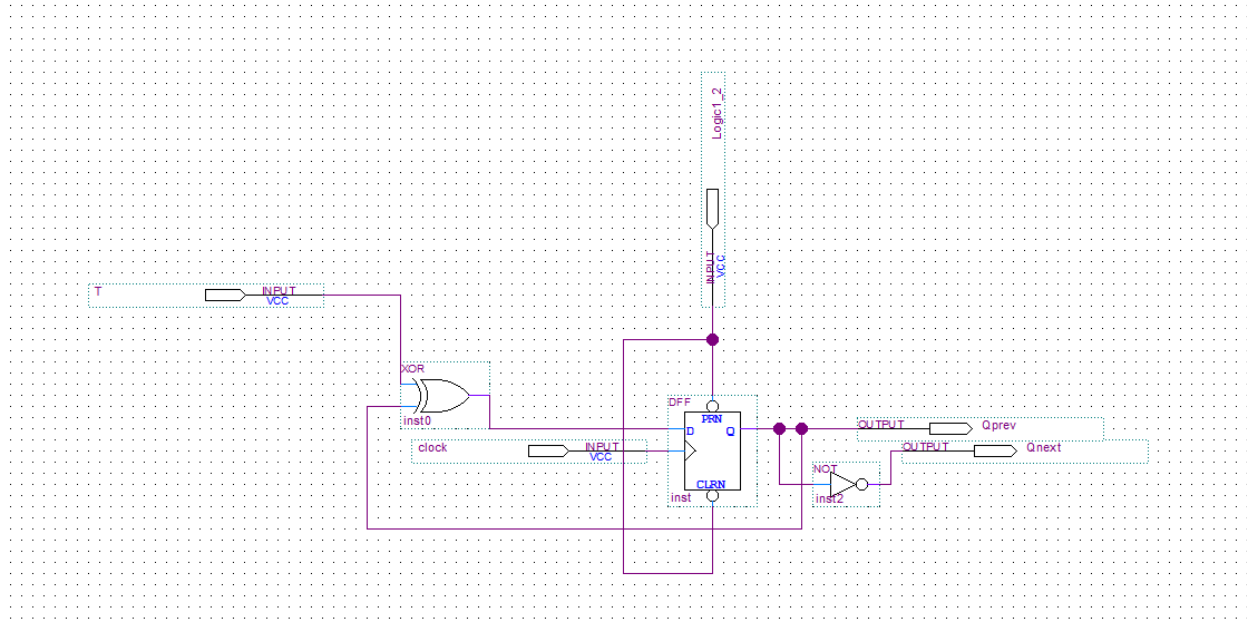


Characteristic Table

Enable	D	Q_+	Q_{++1}
0	x	0	0
0	x	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

Part 3- D Flip-Flop

The third part of the pre-lab examined the D flip-flop. A block diagram of the D flip-flop was created and its characteristic table was written by analyzing the clocked behaviour described in the handout. Unlike the D latch, which is level-sensitive, the D flip-flop changes state only when the clock experiences its triggering edge. The pre-lab implementation was drawn using NAND gates, and particular attention was given to how the circuit samples the input D and transfers that value to the output only at the clock transition.



T	Q _{now}	Q _{next}	Description:
0	0	0	no change
0	1	1	no change
1	0	1	Toggle
1	1	0	toggle.

$Q_{next} = T \oplus Q_{now}$

Excitation Table:

Equipment

Hardware

- Altera DE2-115 Development and Education Board
Used to implement and test sequential logic circuits such as latches and flip-flops. The board switches and push buttons were used as logic inputs, while LEDs were used to display the output states Q and Q'.
- USB-Blaster Programming Cable

Software

- Intel Quartus Prime
Used to design the circuits using the Block Diagram/Schematic Editor, compile the projects, simulate the behaviour using waveforms, and program the board.
- Quartus Block Diagram/Schematic Editor
Used to draw the SR latch, D latch, D flip-flop, and T flip-flop circuits.
- Quartus Simulation Waveform Editor
Used to verify the timing behaviour of the latch and flip-flop outputs.

Logic Components

- NAND gates
- NOR gates
- NOT gates
- Clocked latch and flip-flop subcircuits created during earlier parts of the lab

Input and Output Devices

- Toggle switches on the DE2-115 board for logic inputs such as S, R, D, EN, and T
- Push buttons or assigned clock input for the clocked circuits
- LEDs on the DE2-115 board for observing Q and Q'

Additional Equipment

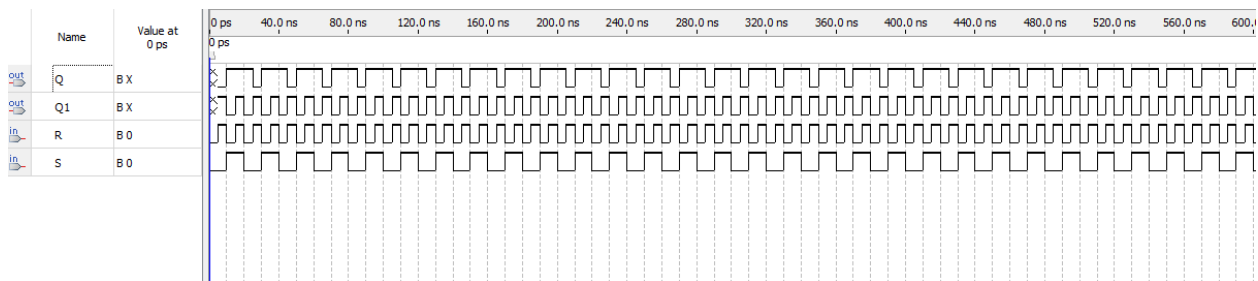
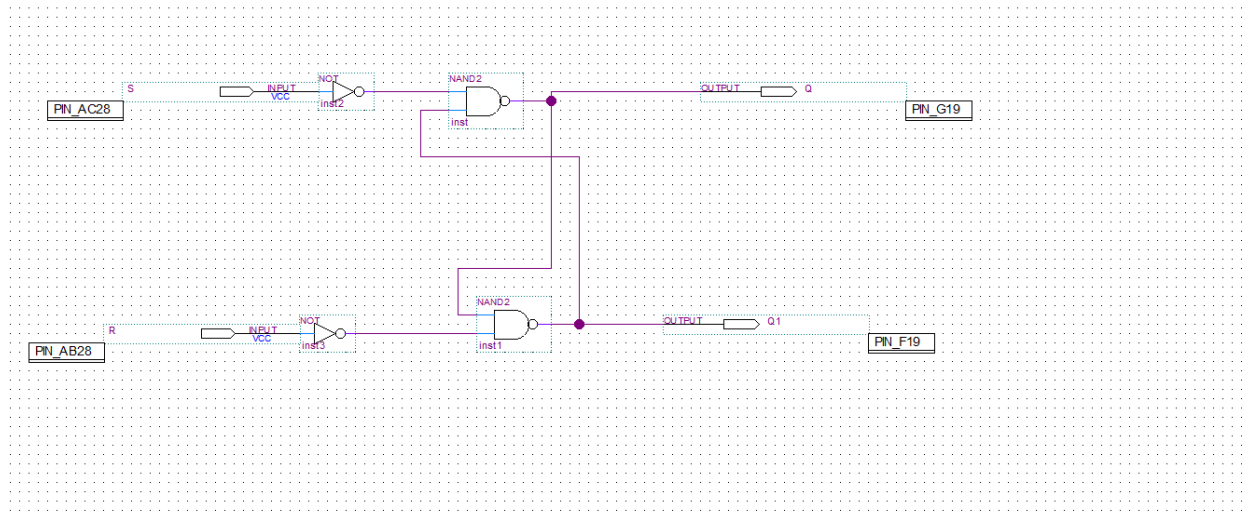
- Laboratory computer capable of running Quartus
- USB connection cable for the board

Design Procedure

The purpose of this experiment was to design and test several sequential logic circuits using Quartus and the Altera board. The circuits prepared during the pre-lab were recreated using the Block Diagram/Schematic Editor, compiled, and then verified through simulation waveforms. After simulation, the input and output pins were assigned to switches, push buttons, and LEDs on the development board so that the circuits could be tested physically. The measured behaviour of each circuit was then compared against the expected theoretical characteristics.

Part 1 – SR Latch

The first in-lab circuit implemented was the NAND SR latch. The circuit was created in Quartus by connecting two NAND gates in a cross-coupled arrangement to form a memory element with complementary outputs Q and Q' . After compiling the project, different input combinations were applied to experimentally determine the function table of the latch, that is, the next state Q_{t+1} as a function of the present inputs and current state. The design was then assigned to the appropriate board inputs and outputs and demonstrated on the DE2-115 board.



Experimental Truth table:

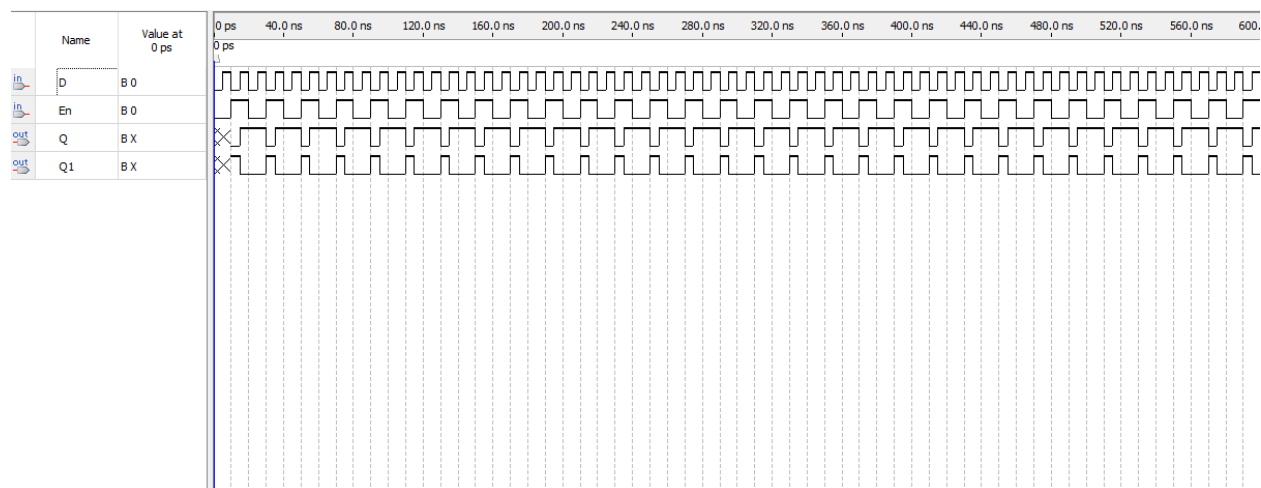
	S	R	Expected Q
1. Set	1	0	1
2. Hold	0	0	1
3. Reset	0	1	0
4. Hold	0	0	0
5. Invalid	1	1	1

Circuit Logic:

In the hold condition, the output remains equal to the previous state, in the set condition, the output is forced high, in the reset condition, the output is forced low, in the forbidden condition, the output becomes invalid or depends on the implementation of the latch.

Part 2 – D Latch

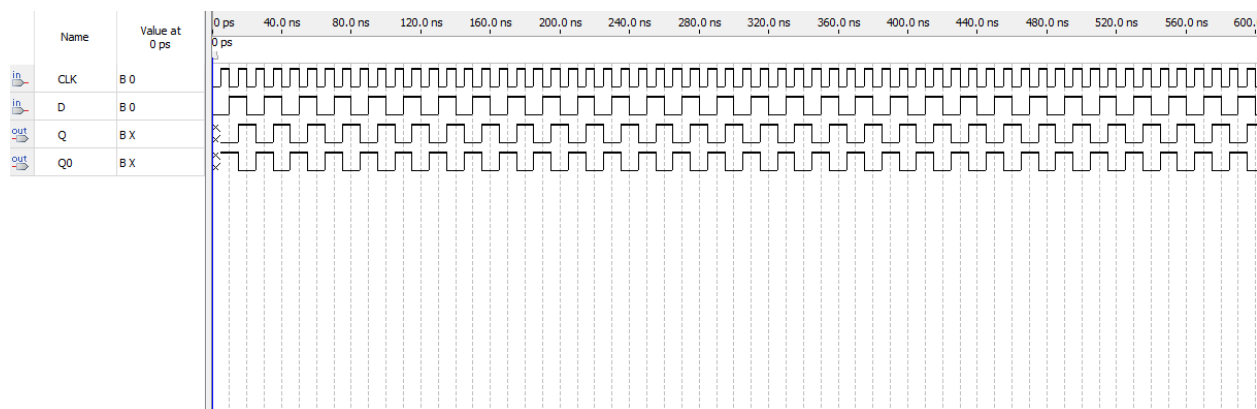
The diagram shows a logic circuit for a 2-bit counter. It consists of two 3-input NAND gates (NAND2) and two D flip-flops (DFF). The inputs to the NAND gates are the outputs of the two D flip-flops, labeled Q and Q1. The outputs of the NAND gates are connected to the D inputs of the flip-flops. The flip-flops are labeled PIN_G19 and PIN_F19. The circuit is implemented using two 3-input NAND gates (NAND2) and two D flip-flops (DFF). The inputs to the NAND gates are the outputs of the two D flip-flops, labeled Q and Q1. The outputs of the NAND gates are connected to the D inputs of the flip-flops. The flip-flops are labeled PIN_G19 and PIN_F19.



A	B	C	D	E	F	G	H
En	D	Qnext	Q1next (Q ⁺)	Mode	Description		
0	X	Qprev	Q1prev	Hold	Changes to D are ignored; the latch maintains its state.		
1	0	0	1	Reset	Output Q follows D to a logic low state.		
1	1	1	0	Set	Output Q follows D to a logic high state.		

When $EN=1$ and $D=0$, the output is reset, when $EN=1$ and $D=1$, the output is set, when $EN=0$, the output stores the previous state regardless of changes at D .

Part 3 – D Flip-Flop

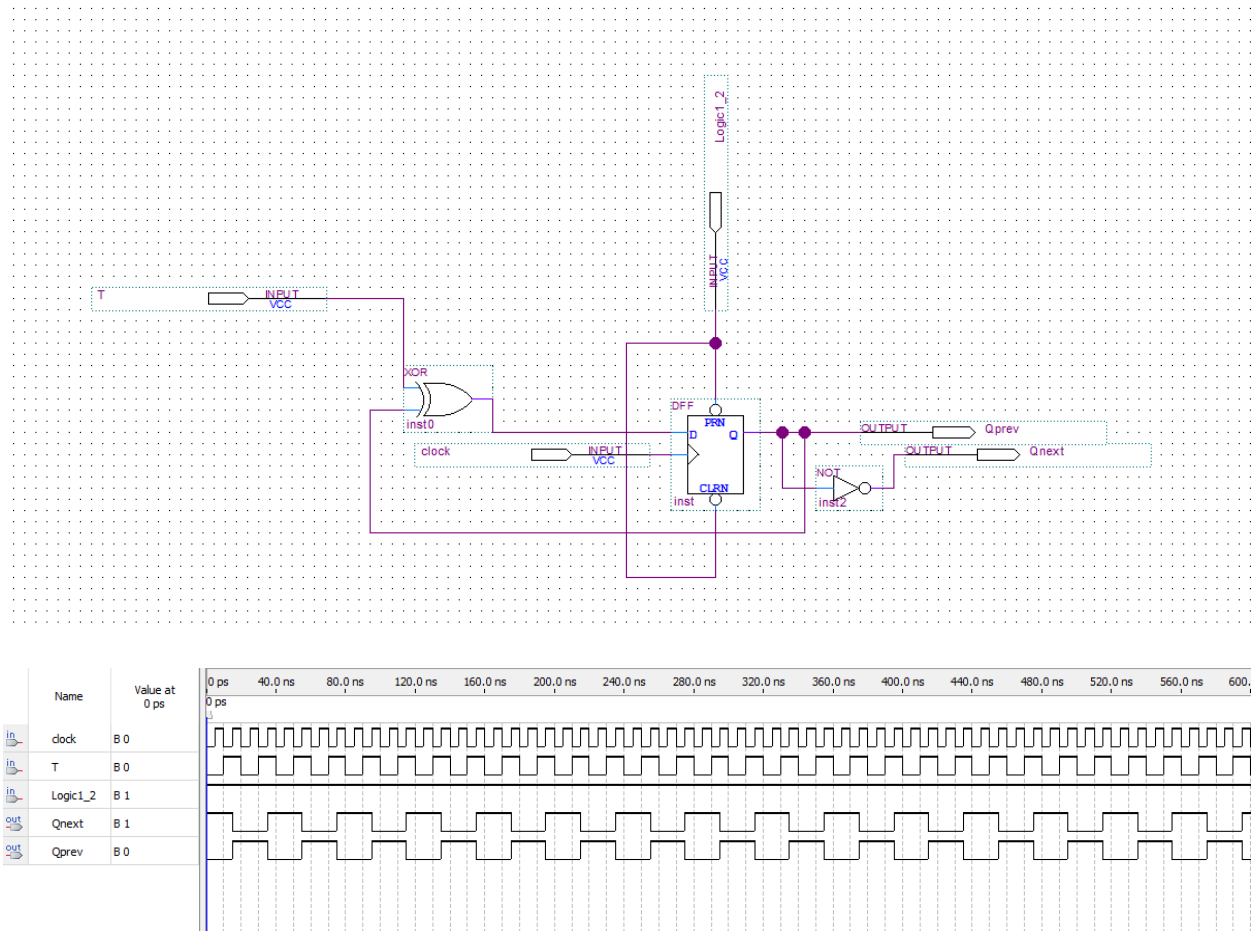


Step	CLK (Input)	D (Input)	Q (Output)	Mode	Observation
1		0	0 Qprev	Hold	Changing D has no effect on the LED.
2		0	1 Qprev	Hold	State remains latched from previous step.
3		1	0	0 Reset	Output Q immediately follows D to 0.
4		1	1	1 Set	Output Q immediately follows D to 1.
5	1 → 0		1	1 Latch	CLK drops; the '1' is now "trapped" or stored.
6		0	0	1 Hold	D changed to 0, but Q stays 1 because CLK is 0.

On the active clock edge, when $D=0$, the next state becomes 0, on the active clock edge, when $D=1$, the next state becomes 1, at times other than the triggering clock edge, the output retains its previous state.

Part 4 – T Flip-Flop

The final circuit implemented was the T flip-flop. This circuit was constructed using the D flip-flop designed in Part III together with the required feedback logic so that the next state depended on both the present state and the toggle input T . After drawing the circuit and compiling the project, the built T flip-flop was tested to experimentally determine its behaviour. The design was then connected to the board inputs and outputs and demonstrated on the board for evaluation.



Circuit Logic:

When $T=0$, the output remains in its previous state on the clock event, when $T=1$, the output toggles to the opposite state on each active clock event.

Discussion of Results

The experimental results obtained from both simulation and hardware implementation confirmed the theoretical behavior of the sequential logic circuits analyzed in this lab. For the SR latch, the observed outputs matched the expected hold, reset, and set conditions for the allowed input

combinations, while the forbidden SR condition was identified as an invalid operating state. The D latch results also agreed with theory, since the circuit removed the forbidden state of the SR latch and operated in two distinct modes: when EN was active, the output followed the data input D, and when EN was inactive, the latch stored its previous value. This showed that the circuit was capable of storing state information rather than responding only to the present input values.

Waveforms generated in Quartus again served as a verification tool for comparing theoretical predictions with the actual behavior of the circuits. The D flip-flop waveform showed that the circuit was edge-sensitive and that the input D was sampled on the positive edge of the clock, while changes occurring when the clock was stable or going low had no effect on the output. The T flip-flop results also matched the expected behavior, since the output toggled between HIGH and LOW on each clock pulse when T was active, and remained unchanged when T was inactive. These results clearly demonstrated the difference between level-sensitive latch behavior and edge-triggered flip-flop operation.

Overall, there was excellent agreement between the theoretical tables, simulation waveforms, and the hardware tests performed on the Altera DE2-115 board. No major discrepancies were observed, and the results demonstrated that the circuits were accurately designed and correctly implemented using basic logic gates. This lab therefore provided a practical demonstration of how sequential logic circuits store information and respond over time, which is one of the fundamental differences between sequential and combinational digital systems.

Conclusion

This lab focused on the analysis, implementation, and testing of several important sequential logic circuits, namely the SR latch, D latch, D flip-flop, and T flip-flop. By constructing these devices from basic gates, deriving their characteristic and excitation tables, and observing their responses through simulation and hardware testing, the experiment successfully illustrated the principles of latching, memory storage, and edge-triggering. The results obtained in Quartus and on the board showed that each circuit behaved according to the function tables provided in the lab instructions.

The experimental results showed no significant differences between the expected theoretical behavior and the observed outcomes. The SR latch correctly demonstrated the set, reset, and hold states, the D latch followed the data input only when enabled, the D flip-flop responded on the clock's positive edge, and the T flip-flop toggled as expected on successive clock pulses. Together, these results reinforced the importance of latches and flip-flops as fundamental building blocks of sequential logic systems and demonstrated how timing and stored state play a central role in digital circuit design.

Appendices

Appendix A - Prelab (Screenshots and Tables)

Part 1 – SR Latch

- SR Latch Circuit Diagram (Prelab)
- SR Latch Simulation Waveform (Prelab)

Part 2 – D Latch

- D Latch Circuit Diagram (Prelab)
- D Latch Simulation Waveform (Prelab)
- D Latch Characteristic Table (Prelab)
- D Latch Experimental Observation Table (Prelab)

Part 3 – D Flip-Flop

- D Flip-Flop Circuit Diagram (Prelab)
- D Flip-Flop Simulation Waveform (Prelab)
- D Flip-Flop Characteristic Table (Prelab)

Part 4 – T Flip-Flop

- T Flip-Flop Circuit Diagram (Prelab)
- T Flip-Flop Simulation Waveform (Prelab)

Appendix B - In Lab

- Sr_Latch.bdf
- Sr_Latch.qpf
- Sr_Latch.qsf

- D_Latch.bdf
- D_Latch.qpf
- D_Latch.qsf

- D_FlipFlop.bdf
- D_FlipFlop.bsf
- D_FlipFlop.qpf
- D_FlipFlop.qsf
- D_FlipFlop.qws

- Tflipflop.bdf
- Tflipflop.qpf
- Tflipflop.qsf
- Tflipflop.qws

Appendix C – Simulation and Verification Files

- Waveform.vwf (Part I – SR Latch)
- Waveform1.vwf (Part I – SR Latch)
- Waveform1.vwf.temp (Part I – SR Latch)
- Sr_Latch_nativelink_simulation.rpt
- Waveform.vwf (Part II – D Latch)
- Waveform.vwf.temp (Part II – D Latch)
- Waveform.vwf (Part IV – T Flip-Flop)

Appendix D – Screenshots (Circuits, Waveforms, Tables)

Part 1 – SR Latch

- SR Latch Circuit Diagram
- SR Latch Simulation Waveform

Part 2 – D Latch

- D Latch Circuit Diagram
- D Latch Simulation Waveform
- Experimental Function Table (Part II)

Part 3 – D Flip-Flop

- D Flip-Flop Circuit Diagram
- D Flip-Flop Simulation Waveform
- Experimental Function Table (Part III)

Part 4 – T Flip-Flop

- T Flip-Flop Circuit Diagram
- T Flip-Flop Simulation Waveform

Appendix E – Software Environment

- Intel Quartus Prime
- Quartus Block Diagram/Schematic Editor
- Quartus Simulation Waveform Editor